

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1

Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I S.Rathika(192419042) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Rathika

Logical Agents – Case Study Solutions

Propositional Logic, FOL, Forward/Backward Chaining, Unification, Resolution, and Wumpus World

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Define the propositional symbols:

- F_A : Patient A has Fever
- C_A : Patient A has Cough
- B_A : Patient A has Breathlessness
- $Rash_A$: Patient A has Rash
- Flu_A : Patient A has Possible Flu
- M_A : Patient A has Possible Measles
- P_A : Patient A has Possible Pneumonia

Rules:

Rule I: $F_A \wedge C_A \rightarrow Flu_A$

Rule II: $F_A \wedge Rash_A \rightarrow M_A$

Rule III: $C_A \wedge B_A \rightarrow P_A$

Equivalent CNF forms:

$\neg F_A \vee \neg C_A \vee Flu_A$

$\neg F_A \vee \neg Rash_A \vee M_A$

$\neg C_A \vee \neg B_A \vee P_A$

Facts:

F_A, C_A, B_A

Therefore, the knowledge base contains the three rules together with the three facts above.

(b) Forward Chaining

Initial facts: F_A, C_A, B_A .

1. Rule I fires because F_A and C_A are true. Derive Flu_A (Possible Flu).
2. Rule III fires because C_A and B_A are true. Derive P_A (Possible Pneumonia).
3. Rule II cannot fire because $Rash_A$ is not given.

Final forward-chaining conclusions:

- Possible Flu
- Possible Pneumonia

Possible Measles is not derivable from the given facts.

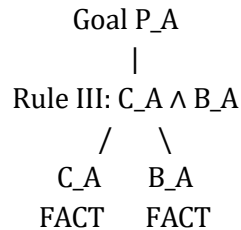
(c) Backward Chaining for Pneumonia

Goal: P_A (Patient A has Possible Pneumonia). Rule III has P_A as its conclusion:

$$C_A \wedge B_A \rightarrow P_A$$

The goal is reduced to two subgoals: C_A and B_A. Both are explicitly present as facts.

Goal reduction tree:



Therefore $KB \models P_A$. Backward chaining confirms Possible Pneumonia.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification

Rule 1:

$$\forall x [Vehicle(x) \wedge EmergencyType(x) \rightarrow ClearPath(x)]$$

Given facts: Vehicle(Ambulance) and EmergencyType(Ambulance).

$$MGU \theta_1 = \{x/Ambulance\}$$

$$Vehicle(x)\theta_1 = Vehicle(Ambulance)$$

$$EmergencyType(x)\theta_1 = EmergencyType(Ambulance)$$

$$\text{Therefore: } ClearPath(Ambulance)$$

For Rule 2, the same substitution gives GreenSignal(Ambulance) and $\neg RedSignal(Ambulance)$.

For Rule 3, using Vehicle(CarA), Behind(CarA, Ambulance), and GreenSignal(Ambulance):

$$MGU \theta_3 = \{x/CarA, y/Ambulance\}$$

Therefore: Proceed(CarA)

(b) Resolution Proof

CNF clauses:

- Rule 1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
- Rule 2a: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
- Rule 2b: $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$
- Rule 3: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts:

$\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}),$
 $\text{Behind}(\text{CarA}, \text{Ambulance})$

Resolution steps:

4. Resolve Rule 1 with $\text{Vehicle}(\text{Ambulance})$, using $x/\text{Ambulance}$:
 $\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$.
5. Resolve with $\text{EmergencyType}(\text{Ambulance})$: $\text{ClearPath}(\text{Ambulance})$.
6. Resolve Rule 2a with $\text{ClearPath}(\text{Ambulance})$: $\text{GreenSignal}(\text{Ambulance})$.
7. Instantiate Rule 3 with $\{x/\text{CarA}, y/\text{Ambulance}\}$.
8. Resolve with $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$, and $\text{GreenSignal}(\text{Ambulance})$.
9. Derive $\text{Proceed}(\text{CarA})$.

Therefore: KB $\models$ Proceed(CarA)

(c) Two Simultaneous Emergency Vehicles

The universally quantified rules can apply independently to multiple emergency vehicles. For example, if FireTruck is also an emergency vehicle and the facts $\text{Vehicle}(\text{FireTruck})$ and $\text{EmergencyType}(\text{FireTruck})$ are added, Rule 1 derives $\text{ClearPath}(\text{FireTruck})$, and Rule 2 derives $\text{GreenSignal}(\text{FireTruck})$.

However, realistic simultaneous-emergency management requires additional facts and rules, such as:

- Position and route facts for every emergency vehicle.
- Conflict detection rules for vehicles requesting the same intersection.
- Priority rules to determine which emergency vehicle receives precedence.
- Collision-avoidance or holding rules when simultaneous green signals would conflict.

Thus the KB is logically general enough to represent multiple emergency vehicles, but it is not sufficient for complete conflict-aware traffic management.

Case Study 3 — Agricultural Expert Reasoning System

(a) CNF Conversion

Let S = SoilDry, I = IrrigationNeeded, W = CropWheat, D = ApplyDripMethod, and R = CropAtRisk.

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

$$S \rightarrow I \equiv \neg S \vee I$$

$$\text{CNF: } \neg S \vee I$$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

$$(I \wedge W) \rightarrow D$$

$$\equiv \neg(I \wedge W) \vee D$$

$$\equiv \neg I \vee \neg W \vee D$$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

$$(\neg D \wedge W) \rightarrow R$$

$$\equiv \neg(\neg D \wedge W) \vee R$$

$$\equiv D \vee \neg W \vee R$$

Facts:

S

W

Complete CNF:

$$(\neg S \vee I) \wedge (\neg I \vee \neg W \vee D) \wedge (D \vee \neg W \vee R) \wedge S \wedge W$$

(b) Resolution Refutation of ApplyDripMethod

Goal: D . Negate the goal and add $\neg D$ to the CNF knowledge base.

Relevant clauses:

10. $\neg S \vee I$

11. $\neg I \vee \neg W \vee D$

12. S

13. W

14. $\neg D$ (negated goal)

Resolution:

15. Resolve $\neg S \vee I$ with S to derive I .
16. Resolve $\neg I \vee \neg W \vee D$ with I to derive $\neg W \vee D$.
17. Resolve $\neg W \vee D$ with W to derive D .
18. Resolve D with the negated goal $\neg D$ to derive the empty clause $\square$.

$$KB \wedge \neg D \vdash \square$$

Therefore $KB \models \text{ApplyDripMethod}$.

(c) Removing SoilDry

If **SoilDry** is removed, only **CropWheat** remains as an explicit fact.

19. Rule P1 cannot fire because **SoilDry** is unavailable, so **IrrigationNeeded** is not derivable.
20. Rule P2 cannot fire because **IrrigationNeeded** is unavailable, so **ApplyDripMethod** is not derivable.
21. Rule P3 requires $\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$. Although **ApplyDripMethod** cannot be proved, its negation is also not given. Therefore **CropAtRisk** is not derivable.

Final status:

- **CropWheat**: known
- **IrrigationNeeded**: not derivable
- **ApplyDripMethod**: not derivable
- **CropAtRisk**: not derivable

This illustrates the open-world principle: failure to prove a proposition does not prove its negation.

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

Initial percepts:

$$\text{Stench}[1,2], \text{Breeze}[1,1], \text{Glitter}[2,2]$$

Rule 5: $\text{Stench}[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$. By Modus Ponens:

$$\text{WumpusAdjacent}[1,2]$$

Rule 6: $\text{Breeze}[1,1] \rightarrow \text{PitAdjacent}[1,1]$. By Modus Ponens:

$$\text{PitAdjacent}[1,1]$$

Rule 7: $\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$. With $x=2, y=2$:

$$\text{Gold}[2,2]$$

Rule 8 requires explicit $\neg \text{Wumpus}[x,y]$ and $\neg \text{Pit}[x,y]$. No such negative facts are provided, so no $\text{Safe}[x,y]$ fact can be derived.

Derived conclusions:

- Wumpus is adjacent to [1,2].
- A pit is adjacent to [1,1].
- Gold is at [2,2].
- No cell is proven safe by Rule 8.

(b) Forward Chaining: Possible Wumpus and Pit Locations

Using standard four-directional adjacency:

Neighbors of [1,2] are [1,1], [1,3], and [2,2]. Therefore the Wumpus may be at:

$$\text{Wumpus} \in \{[1,1], [1,3], [2,2]\}$$

Neighbors of [1,1] are [1,2] and [2,1]. Therefore the pit may be at:

$$\text{Pit} \in \{[1,2], [2,1]\}$$

The percepts identify possible locations, not a unique Wumpus or pit location.

(c) Safety of Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

At [1,2], the Breeze at [1,1] means a pit may be at [1,2] or [2,1]. Hence [1,2] is not proven safe.

At [2,2], the Stench at [1,2] means the Wumpus may be at [2,2]. Therefore [2,2] is also not proven safe, even though Gold[2,2] is known.

At [1,1], there is no explicit proof of a pit or Wumpus, but Rule 8 still cannot prove safety because explicit negative facts are absent.

Final conclusion: the complete path is NOT logically guaranteed to be safe under the supplied knowledge base. The agent should not classify the route as safe without additional percepts or negative knowledge.

Overall Conclusions

Case Study	Main Conclusion
1	Patient A has Possible Flu and Possible Pneumonia; backward chaining confirms Pneumonia.
2	Proceed(CarA) follows by unification and resolution.
3	ApplyDripMethod is proven; without SoilDry, irrigation, drip application, and CropAtRisk are not derivable.

4	Gold is at [2,2], but the proposed path is not guaranteed safe because [1,2] may contain a pit and [2,2] may contain the Wumpus.
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